

MAME LIBASSE MBOUP

Data / AI Engineer

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5+ years of experience in **data engineering**, **data science** and **machine learning**, in **banking** (BPCE Group), **public health** (ARS PACA / Ségur) in **HDS-certified** and **GDPR-compliant** environments, **telecommunications** (Orange) and **energy** (Enedis). Expertise in **Agentic AI**, Data Analysis, Machine Learning, BI and ELT. Proficient in **Python**, **SQL**, **GCP/BigQuery**, **dbt**, **Power BI** and **n8n**.

PROFESSIONAL EXPERIENCE

Data Scientist – Data Lab & New Use Cases Squad BPCE-SI (BPCE Group)

📅 Jul. 2026 – Present 📍 Aix-en-Provence, France

- Performance & Run team: ensuring stability of existing data systems, incident management and delivery of enhancements, working directly with BPCE-SI business units.
- Development and industrialization of data pipelines on Google Cloud Platform (BigQuery, Cloud Composer, Cloud Storage) for the Group Cloud Data Platform.
- Analytics modeling and ELT transformations with dbt (staging/intermediate/marts layers, data quality tests, model documentation and lineage).
- Design and industrialization of Power BI reports on top of BigQuery (semantic modeling, DAX, RLS security) for business steering.
- Generative AI use cases: LLM integration on Group data (analytics assistants, automated document processing) within the Data Lab.
- Agent-assisted development practice in a regulated banking environment (internal data, compliance constraints): structured context documentation of repositories and accelerated onboarding onto legacy codebases.

Data Scientist – Public Health Innovation e-Santé Sud (GRADEs PACA)

📅 Oct. 2022 – Jul. 2026 📍 Hyères, France

- Hospital bed mapping: real-time visualization tool for available healthcare capacities, powered by automated data feeds, deployed for crisis management (Covid-19, mass casualty plans, hospital strain) at ARS PACA.
- CONTESS: decision-support dashboard for the Ségur digital health program across 6 sectors, cross-referencing DMP, MSS, INS and Pro Santé Connect usage via multi-source SAS Viya pipelines.
- Strategic dashboards for regional management and care coordination (OKAPI, ETP, Emergency/SAMU-SMUR, ROR, E-parcours).
- TRU Modeling (ARS PACA): identifying at municipal level the factors influencing avoidable emergency visits (demographic, health, care access and socioeconomic variables).
- Migration of BI reports from Jaspersoft to SAS Viya, Python/SQL automation and regional SAS Viya platform administration (access rights, Caslibs, user groups).
- Automated medical data extraction (reports, prescriptions, lab results in PDF/images) via multimodal LLM (Gemini), with automatic structuring usable through n8n.
- Healthcare professional address verification via cross-checking (Google Maps/Apify, PagesJaunes/SerpAPI) and intelligent Redmine ticket triage through LLM analysis of business content.

Data Scientist Orange

📅 Sept. 2021 – Sept. 2022 📍 Lyon, France

- End-to-end processing of fibre-optic rollout data, from customer order through to connection, for lead-time tracking and bottleneck identification.
- RPA automation (UI Path) and Kibana / Power BI dashboards for operational steering of rollout teams.
- Development of statistical models on large-scale heterogeneous data streams (network monitoring).

Data Engineer Enedis

📅 Apr. 2021 – Sept. 2021 📍 Toulon, France

- Full-stack regional energy monitoring application (CodeIgniter/PHP/MySQL) for real-time field consumption tracking.
- Multi-source data integration via Denodo (VQL), analytical reports for regional management.

PROJECTS & RESEARCH

Academic Research, ML & Deep Learning

- Co-clustering of spatio-temporal data in R — M2 Thesis, Université Lyon 2. Paper on article categorization via semantic keyword relationships.
- Word Embedding, Topic Modeling (LDA), DBLP visualization (NetworkX, Umap, Pyvis), Skip-Gram (GloVe), RNN, CNN, Encoder-Decoder + Attention, fine-tuning BERT. Random Forests from scratch (OOP), mixture models (EM, SVM, K-Means) on MNIST.

Full-Stack AI Platforms

- LibJobPilot: job search automation — scraping (LinkedIn/Indeed/Glassdoor), AI scoring of postings, auto-apply via Playwright, Kanban dashboard (React, n8n, Supabase, LiteLLM).
- LibLeadIN: B2B prospecting with AI agents — prospect enrichment, personalized messaging, automated sending and tracking (Next.js, n8n, DeepSeek).
- LiBrain: multi-agent assistant (emails, calendar, contacts) — n8n orchestrator, 4 connected agents, multi-provider LiteLLM proxy.
- LibOrga: mobile-first productivity app (Eisenhower matrix) — natural-language task capture, OCR of handwritten lists (tesseract.js), installable PWA. Next.js 16, React 19, Supabase (Postgres, RLS).
- LibCalPilot: nutrition-tracking PWA for athletes — AI multi-food photo recognition (Gemini vision), barcode scanning, CIQUAL/OpenFoodFacts database, requirement calculation (Mifflin-St Jeor). React 19, Supabase Edge Functions, TDD.

EDUCATION

Master Machine Learning for AI Université Lumière Lyon 2

📅 Sept. 2020 – Sept. 2022

- Advanced supervised learning, Deep learning, Manifold learning, Model-based learning, Parallel computing, NLP, Text Mining, Big Data management and analytics.

BSc Computer Science Université du Littoral Côte d'Opale

📅 Sept. 2018 – Sept. 2020 · Calais, France

BSc Mathematics & Computer Science Université de Thiès

📅 Sept. 2015 – Sept. 2018 · Thiès, Senegal

SKILLS

► Programming & Dev

Python SQL R SAS Bash Git Bitbucket
REST API React / Next.js Docker CI/CD

► Machine Learning & Deep Learning

Scikit-learn TensorFlow PyTorch NLP Time Series
Clustering Classification Regression

► Data Engineering & Cloud

PostgreSQL MongoDB Hadoop dbt Spark
GCP BigQuery Cloud Composer Supabase ETL
Airflow

► BI & Automation

SAS Viya Power BI n8n UI Path (RPA) Streamlit
HDS GDPR

► Generative AI & Agents

LLM RAG Fine-tuning Prompt Engineering
AI Agent Claude Code OpenClaw LangChain
LiteLLM Playwright AI Engineering

LANGUAGES

French ●●●●●●
Wolof ●●●●●●
English ●●●●●●
German ●●●●●●

CONTINUING EDUCATION

Coursera Specializations: MLOps, AI Agents, Prompt Engineering, Generative AI, Healthcare AI, Data Modeling & Transformation.

INTERESTS

- **Sports** Weightlifting, Football
- **Entrepreneurship** AI & Health projects
- **Video Games** FC, GTA